Housing Policy, Portfolio Risk, and the Home-Purchase Threshold

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Outline

Recap

Quantitative Implementation

Recap

Recap I

Previously on last presentation

- Research question:
  Does endogenous portfolio reoptimization change the equilibrium transmission of housing policy?

$$\boxed{\text{Housing policy} \;\rightarrow\; x^* \;\rightarrow\; \Pr(a' \geq dP) \;\rightarrow\; D^H \;\rightarrow\; P^H}$$

- Two-period toy model
  - Date 0: portfolio choice

$$x^*(a;P) = \arg\max_{x\in[0,1]} E_R \underbrace{[V_1(a'(R),P)]}_{\text{best attainable value of next-period}}$$

  - Date 1: decision making whether or not to buy home conditional on realized liquid financial wealth a'. Market clears $D^H(P) = S^H(P)$

Recap II

Previously on last presentation

Fixed portfolio choice (No Portfolio Feedback):

$$x = x_0^*(a)$$

Full portfolio choice (Portfolio Feedback):

$$x = x^*(a; P, \text{policy})$$

We can calculate policy amplification ratio

$$\mathcal{A} = \frac{\Delta P^H_{full}}{\Delta P^H_{fixed}}$$

where ΔP represents the house price change after the relevant policy.

Quantitative Implementation

Computation Strategy I

I impose specific functional forms for each value function

$$V_1^R(a', P) = u(a' + y) = \frac{(a' + y)^{1-\gamma} - 1}{1 - \gamma} + \epsilon_R,$$
$$V_1^H(a', P) = u(a' + y - (d + \chi)P) + \psi + \epsilon_H$$

where ψ: utility premium of home ownership, χ: home user cost, ϵ_R, ϵ_H: preference shock for rent, and home purchase.

Assuming $\epsilon_R, \epsilon_H \sim \text{EV1}(0, \sigma_h)$,

$$\pi_H(a', P) = \Lambda\left(\frac{V_1^H(a', P) - V_1^R(a', P)}{\sigma_h}\right)$$

where $\Lambda(z) = \dfrac{1}{1 + e^{-z}}$, logistic CDF.

Computation Strategy II

For housing supply curve, I consider

$$S^H(P) = \bar{H}\left(\frac{P}{\bar{P}}\right)^{\eta}$$

where $\bar{P}$: reference price, $\bar{H}$: reference housing supply at $\bar{P}$.

Parameters and Calibration I

Parameter	**Value**	**Source and construction**
Initial financial wealth a	$F(a, y)$	2025 SCF: `LIQ + STOCKS + BOND`
Labor income y	$F(a, y)$	2025 SCF: `WAGEINC`; joint household observations
Baseline house price P_0	\$398,917	NAR Sep 2026 existing-home median (\$417,500), converted to 2025 dollars
Down-payment ratio d	0.10	NAR 2025 first-time buyers' median down payment
Reference house price $\bar{P}$	\$398,917	Normalization; set equal to P_0
Supply intercept $\bar{H}$	0.08803	Calibrated to clear the housing market at P_0
Supply elasticity η	0.40	Assumption

- SCF sample: Ages 25-44 households with no housing assets

Parameters and Calibration II

Parameter	**Value**	**Source**
CRRA risk aversion γ	2.0	Assumption
Home-ownership premium ψ	0.45	Assumption
Housing taste-shock scale σ_h	0.08	Assumption; logit choice heterogeneity
Safe gross return R_f	1.02	Assumption
Risky gross return $\mathbb{E}[R]$	0.983	Assumption
SD(R)	0.7107	Matched to illustrative two-point return moments
Lognormal parameters (μ_R, σ_R)	$(-0.2274, 0.6485)$	Implied by the assumed risky-return mean and SD
Home-user cost χ	0.02	Informed assumption; Fannie Mae's 1–4% annual maintenance-budget guideline

Preliminary Results

We consider policy menu $d : 0.1 \to 0.075, \bar{H} : 1 \to 1.1$.

Scenario	House price	$\bar{x}$	Purchase rate
Baseline	\$398,917	2.913%	8.803%
DPA — Fixed	\$454,502	2.913%	9.275%
DPA — Full	**\$468,531**	**2.960%**	**9.388%**
Supply — Fixed	\$355,890	2.913%	9.251%
Supply — Full	**\$363,316**	**3.136%**	**9.328%**

Policy Discussion

Portfolio-feedback metric	**Value**	**Interpretation**
DPA amplification	**1.252**	Housing price increases more with portfolio feedback.
Supply response ratio	**0.827**	Portfolio response offsets 17.3% of the price decline.

Why does it matter?

- Ignoring portfolio feedback may mislead policy effects, overlook distributional consequences, and potentially misrank alternative housing policy options.
- Worth researching further

What Next?

- Welfare analysis: who wins and loses under each policy alternative?
  - prospective homebuyers vs. incumbent homeowners,
  - comparison within prospective homebuyers across $F(a, y)$.

- Fine-tuning parameters

- Refinement of model

- Research existing or planned housing policy to relate the model

Appendix

Motivation

Young Americans are taking large financial risks.

	Age group		
Investment attitude / behavior	**18–34**	35–54	55+
Need to take big risks to reach financial goals	**62%**	42%	16%
Traded options	**43%**	24%	10%
Purchased meme / viral investments	**29%**	18%	2%
Invested in cryptocurrency[a]	**50%**	36%	10%

[a] Among respondents aware of cryptocurrency.
Source: FINRA Investor Education Foundation, 2024 Investor Survey.
Survey respondents have investments outside of retirement accounts.

- Are they simply risk-lovers?

What if housing market is part of the reason?

The home-purchase wealth hurdle has risen across generations.

	Age 25–34		
Home-purchase hurdle	**1985**	**2000**	**2025**
Median weekly earnings	$349	$549	$1,143
Annualized earnings[a]	$18.1K	$28.5K	$59.4K
Median new-home price	$84.3K	$169.0K	$417.4K
Home price / annualized earnings	4.65×	5.92×	7.02×
20% benchmark down payment	$16.9K	$33.8K	$83.5K
20% down payment / annualized earnings	**93%**	**118%**	**140%**

[a] Median usual weekly earnings multiplied by 52.

Notes: Earnings are for full-time wage and salary workers, ages 25–34, both sexes. Home price is the median sales price of new houses sold. A 20% down payment is an illustrative benchmark, not a universal mortgage requirement.

Sources: U.S. Bureau of Labor Statistics, Current Population Survey; U.S. Census Bureau/HUD, New Residential Sales.



Related Literature

Recent research connects housing attainability to risk-taking behavior.

1. **Housing attainability ⇒ Portfolio risk-taking**
   (Lee and Yoo, 2025; Dai, Savor, and Wilson, 2025; Arkolakis et al., 2026)

2. **Financial wealth/housing policy ⇒ Housing-market equilibrium**
   (Aiello et al., 2023; Abramson and Landvoigt, 2025)

This project:

Housing policy $\Longrightarrow$ Portfolio response $\Longrightarrow$ Housing demand $\Longrightarrow$ House prices

Research Question

- Does endogenous portfolio reoptimization change the equilibrium transmission of housing policy?

Conventional housing policy channel:

$$\text{Housing policy} \rightarrow D^H, S^H \rightarrow \text{P}$$

This paper:

$$\boxed{\text{Housing policy} \;\rightarrow\; x^* \;\rightarrow\; \Pr(a' \geq dP) \;\rightarrow\; D^H \;\rightarrow\; P^H}$$

where x^*: endogenous optimal share invested in the risky asset, a': next-period liquid financial wealth after portfolio returns, d: required down-payment share.

Quantitative Model

Two-Period Prototype Model

Date 0

Date 0: portfolio choice

$a, y \sim F(a, y), P, R \sim F(R)$ known.

$$x^*(a; P) = \arg \max_{x \in [0,1]} E_R \underbrace{[V_1(a'(R), P)]}_{\text{best attainable value of next-period}}$$

where $V_1(a', P)$: continuation value from the optimal housing decision given realized liquid wealth a'.

$$V_1(a', P) = \begin{cases} V_1^R(a', P), & \text{if } a' < dP, \\ \mathbb{E}_\epsilon \max \left\{ V_1^R(a', P), V_1^H(a', P) \right\}, & \text{if } a' \geq dP. \end{cases}$$

where $V_1^R(a', P)$: value of remaining a renter, $V_1^H(a', P)$: value of purchasing the home.

Two-Period Prototype Model

Date 1

Date 1: home-purchase feasibility

Once R realizes, realized liquid wealth is

$$a' = a[(1-x)R_f + xR]$$

where a: initial liquid financial wealth, $x \in [0,1]$: speculative portfolio share, R_f: gross rate of return of safe asset, R: gross rate of return of risky asset with

$$E[R] < R_f.$$

Home purchase is feasible if

$$a' \geq d \times P$$

where d: required down-payment share, P: equilibrium house price.

Portfolio Choice and Housing Demand

Portfolio risk changes the probability of crossing the down-payment threshold.

$$q(a, x; P) = \Pr[a((1-x)R_f + xR) \geq d \times P].$$

Then,

$$\Pr(\text{buy} \mid a) = \underbrace{q(a, x^*(a; P); P)}_{\text{Can buy}} \quad \underbrace{\pi_H(P)}_{\text{Choose to buy}}$$

where $\pi_H(P)$: purchase probability conditional on feasibility.

Aggregate demand:

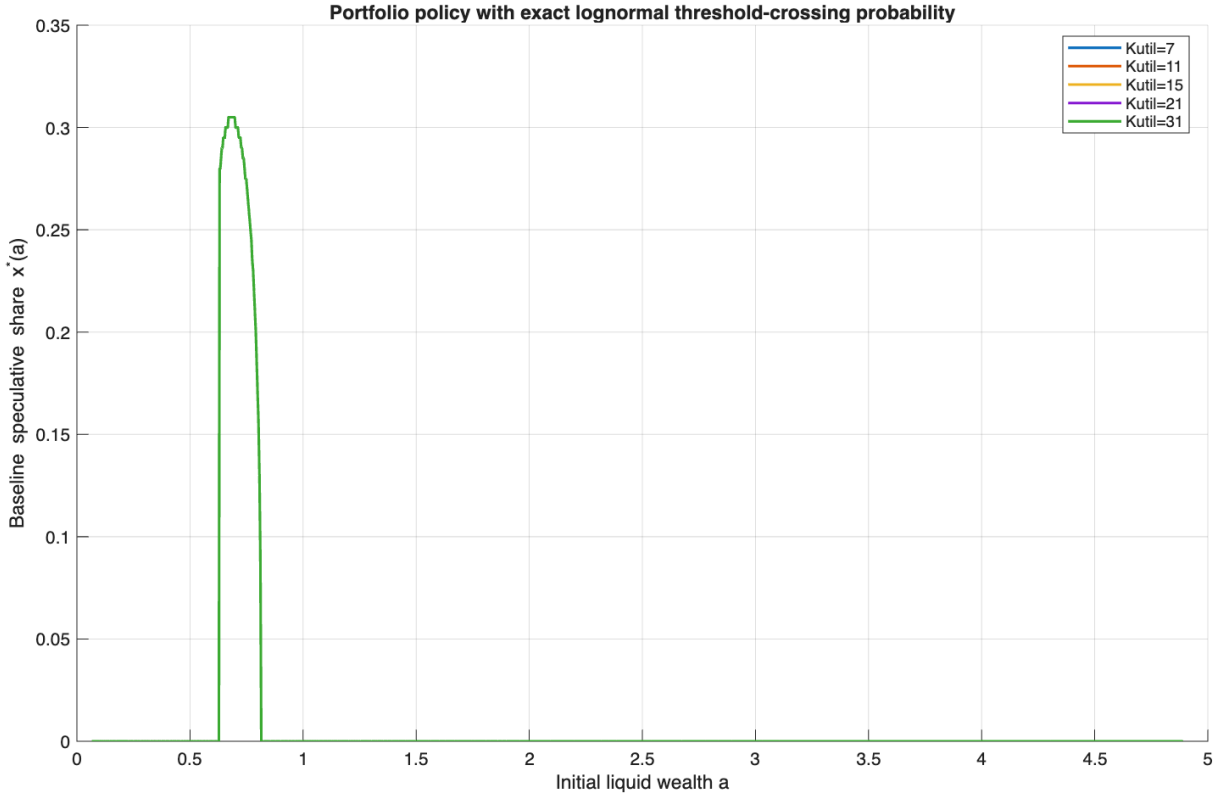
$$D^H(P) = \int q(a, x^*(a; P); P)\pi_H(P)dF(a).$$

In equilibrium,

$$D^H(P) = S^H(P).$$

Home-Purchase Threshold Mechanism

Risk-taking is concentrated near the home-purchase threshold.



Too far	**Within reach**	**Safely attainable**
$x^*(a) = 0$	$x^*(a) > 0$	$x^*(a) = 0$

Identifying the Portfolio-Feedback Effect

Fixed portfolio choice (No Portfolio Feedback):

$$x = x_0^*(a)$$

Full portfolio choice (Portfolio Feedback):

$$x = x^*(a; P, \text{policy})$$

We can calculate policy amplification ratio

$$\mathcal{A} = \frac{\Delta P^H_{full}}{\Delta P^H_{fixed}}$$

where ΔP represents the house price change after the relevant policy.

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